DIPARTIMENTO DI ELETTRONICA, INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

August 30, 2023

Surname:
Name:
Student ID:
Row: Column:

Time: 2 hours Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 30 min before the end of the exam you will get 3 points, if you finish earlier than 20 min you will get 2 points and if you finish earlier than 10 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Describe and compare the Monte Carlo and Temporal Difference approaches to policy evaluation.

Exercise 2 (7 marks)

Describe the role of regularization in managing the bias-variance trade-off. Discuss the difference between L1 and L2 regularization techniques and their respective effects on model complexity and feature selection.

Exercise 3 (2 marks)

Consider the following snippet of code:

```
1  Q = np.zeros(nS * nA)
2  Q_old = np.ones(nS * nA)
3
4  pi = np.zeros((nS, nS * nA))
5  for s in range(nS):
6          pi[s, s*nA:(s+1)*nA] = 1 / nA
7
8  while np.any(Q != Q_old):
9          Q_old = Q
10         Q = (np.eye(nS * nA) - gamma * P_sas @ pi) @ R_sa
11         pi = np.zeros((nS, nS * nA))
12         for s in range(nS):
13                 ga = np.argmax(Q[s*nA:(s+1)*nA])
14                 pi[s, s*nA+ga] = 1
```

1. What algorithm is the snippet of code above implementing? What problem is it trying to solve?
2. Is code snippet above correct? If not, list the mistakes and provide a fix for each of them.
3. Is the `while` loop guaranteed to stop after a finite number of iterations (if `gamma` < 1)? If yes, provide an upper bound on the maximum number of iterations. If not, provide a counterexample.

1. The snippet is implementing the Policy Iteration algorithm. The algorithm is trying to solve the control problem in reinforcement learning.
2. Line 10 is incorrect. The correct line is `np.linalg.inv(np.eye(nS * nA) - gamma * P_sas @ pi) @ R_sa`.
3. Policy Iteration is guaranteed to terminate after a finite number of iterations which is at most the number of deterministic policies nA^{nS}.

Exercise 4 (2 marks)

Assume to have a regression model $y = \mathbf{w}^\top \mathbf{x}$ over a dataset of input $\mathbf{x}_i$ and corresponding target t_i, for $i \in \{1, \dots, N\}$. Indicate whether the following statements are true or false and motivate your answers.

1. Using the Maximum Likelihood Estimation method to compute the optimal parameter vector $\mathbf{w}^*$ has a closed-form solution, regardless of the noise distribution present on the target realizations t_i.
2. The least squares estimator $\mathbf{w}^*$ of the parameter vector $\mathbf{w}$ applied to the given dataset is the one with the lowest Mean Squared Error (MSE) of all linear estimators with no bias.
3. Introducing nonlinear basis functions into the model increases the model flexibility at the cost of reducing its explainability (i.e., its degree of understanding by humans).
4. We can use a regression model even if the target vector $(t_1, \dots, t_N)$ represents a time series (i.e., they are not independent of each other and their distribution depends on the time index i).

1. FALSE: the closed form solution is provided only for specific distribution for the noise. For instance, the one corresponding to the Least Square one requires to have uncorrelated Gaussian noise with fixed variance.
2. TRUE: This is a direct consequence of the Gauss-Markov theorem.
3. TRUE: By adding nonlinear basis we are increasing the dimension of the hypothesis space, therefore increasing the chance that the real model is *close* to one of the model of our hypothesis space. Conversely, it is harder to interpret nonlinear relationships between input and output.
4. FALSE: A main assumption of ML models is to have i.i.d. (independent and identically distributed) data, otherwise we are facing with an environment whose behaviour changes over time or depends on the previous sequence of data. Therefore, even if only the target distribution is not i.i.d. implies that it is not a good idea to use classical ML methods.

Exercise 5 (2 marks)

Indicate whether the following statements are true or false. Justify your answers.

1. A training error that is significantly smaller than the test error is a symptom of underfitting.
2. A training loss close to the test error but significantly larger than the desired loss is a symptom of underfitting.
3. A small-variance model is prone to overfitting.
4. A small-bias model is prone to overfitting.

1. FALSE. It is a symptom of overfitting. Indeed, in such a case, the training error is not representative of the true error. The chosen model has too large variance.
2. TRUE. Indeed, in such a case, the training error is representative of the true error, but the latter is larger than the desired one. The chosen model has too large bias.
3. FALSE. Indeed, the training error will be representative of the true error. Nothing can be stated about the attitude to underfitting.
4. DEPENDS. Nothing can be said about overfitting since it depends on the complexity of the hypothesis space (i.e., the variance of the model). For sure, the model will not be prone to underfitting.

Exercise 6 (2 marks)

You are the owner of a software house and you are willing to estimate how much effort and income you are getting from the development of a new software. Assume there is some historical data about the characteristics of each software developed in the past, the duration of the development, and the profits provided by it.

1. Model the problem of estimating the duration and profits you get from the development of a new software. Specify the data you would use, the class of problem you would model this problem with, and the methodology you would adopt.
2. What if, instead, you do not have a dataset? Assume you need to decide on the sequential choice of the development of new software by choosing from a set of categories (e.g., game, business analytics, finance, sport). Provide a modeling (class of methods and data to process) and a method to solve such an issue.

1. The problem is about *multiple regression* since both the quantity to estimate (amount of time and gains) are ordinal. As an input one might use the details about the software and about the team used to develop it. The solution might be to use the LS to perform multiple linear regression.
2. In this case we are in a *multi-armed bandit* setting, where the option are the different software classes and the reward is about either the gains you get from the development or minus the time duration (taking into account for multiple rewards is not a viable option in the classical MAB setting). One can use either UCB1 or TS to solve the problem.

Exercise 7 (4 marks)

Consider the logistic regression two-class classifier:

$$y_{\mathbf{w}}(\mathbf{x}) = \frac{1}{1+\exp(-\mathbf{w}^\top \mathbf{x})}.$$

1. Given a training dataset $\{(\mathbf{x}_i, t_i)\}_{i=1}^N$, write the formula of the loss function $L(\mathbf{w})$ for training the logistic regression two-class classifier above.
2. Derive the gradient of the loss function $L(\mathbf{w})$.
3. Starting from the weight vector $\mathbf{w}_0 = (0,0,0)^\top$, apply the gradient update with $\alpha = 1$ and the following data points:

$$\mathbf{x}_1 = (1,0,-1)^\top \qquad t_1 = 1,$$
$$\mathbf{x}_1 = (0,2,-3)^\top \qquad t_1 = 0.$$

1. The loss function for training the logistic regression two-class classifier is the negative log-likelihood (a.k.a. cross-entropy):

$$L(\mathbf{w}) = -\log \prod_{i=1}^N y_{\mathbf{w}}(\mathbf{x}_i)^{t_i}(1-y_{\mathbf{w}}(\mathbf{x}_i))^{1-t_i} = -\sum_{i=1}^N \left(t_i \log y_{\mathbf{w}}(\mathbf{x}_i) + (1-t_i)\log(1-y_{\mathbf{w}}(\mathbf{x}_i))\right).$$

2. We first compute the gradient $\nabla_{\mathbf{w}} y_{\mathbf{w}}(\mathbf{x})$:

$$\nabla_{\mathbf{w}} y_{\mathbf{w}}(\mathbf{x}) = \nabla_{\mathbf{w}} \frac{1}{1+\exp(\mathbf{w}^\top \mathbf{x})} = \frac{\exp(-\mathbf{w}^\top \mathbf{x})\mathbf{x}}{(1+\exp(-\mathbf{w}^\top \mathbf{x}))^2} = y_{\mathbf{w}}(\mathbf{x})(1-y_{\mathbf{w}}(\mathbf{x}))\mathbf{x}.$$

Then, the gradient of the loss function can be computed as follows:

$$\begin{aligned}
\nabla_{\mathbf{w}} L(\mathbf{w}) &= -\nabla_{\mathbf{w}} \sum_{i=1}^N \left(t_i \log y_{\mathbf{w}}(\mathbf{x}_i) + (1-t_i)\log(1-y_{\mathbf{w}}(\mathbf{x}_i))\right) \\
&= -\sum_{i=1}^N \left(t_i \frac{y_{\mathbf{w}}(\mathbf{x}_i)(1-y_{\mathbf{w}}(\mathbf{x}_i))\mathbf{x}_i}{y_{\mathbf{w}}(\mathbf{x}_i)} - (1-t_i)\frac{y_{\mathbf{w}}(\mathbf{x}_i)(1-y_{\mathbf{w}}(\mathbf{x}_i))\mathbf{x}_i}{1-y_{\mathbf{w}}(\mathbf{x}_i)} \right) \\
&= -\sum_{i=1}^N \left(t_i(1-y_{\mathbf{w}}(\mathbf{x}_i))\mathbf{x}_i - (1-t_i)y_{\mathbf{w}}(\mathbf{x}_i)\mathbf{x}_i\right) = \sum_{i=1}^N (y_{\mathbf{w}}(\mathbf{x}_i) - t_i)\mathbf{x}_i
\end{aligned}$$

3. Since $\mathbf{w}_0 = (0,0,0)^\top$ we have that $y_{\mathbf{w}_0}(\mathbf{x}) = 1/2$ regardless the value of $\mathbf{x}$. Thus, we have:

$$\begin{aligned}
&\mathbf{w}_1 = \mathbf{w}_0 - \alpha \sum_{i=1}^2 (y_{\mathbf{w}}(\mathbf{x}_i) - t_i)\mathbf{x}_i \\
&= \begin{pmatrix}0\\0\\0\end{pmatrix} - 1\left(\frac{1}{2}-1\right)\begin{pmatrix}1\\0\\-1\end{pmatrix} - 1\left(\frac{1}{2}-0\right)\begin{pmatrix}0\\2\\-3\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix} + \begin{pmatrix}1/2\\0\\-1/2\end{pmatrix} + \begin{pmatrix}0\\-1\\3/2\end{pmatrix} = \begin{pmatrix}1/2\\-1\\1\end{pmatrix}.
\end{aligned}$$

Exercise 8 (4 marks)

Consider the bounds in Figure 1 (supposed to hold with probability at least δ) for a stochastic **Multi–Armed Bandit** (MAB) setting, where the bars are the Hoeffding bounds for the expected rewards, the circles are the estimated expected rewards, and the crosses are the real expected rewards (respectively equal to 0.5, 0.2, 0.1, 0.0, 0.2, 0.4, 0.6, 0.1, 0.1, and 0.3 for the ten arms).

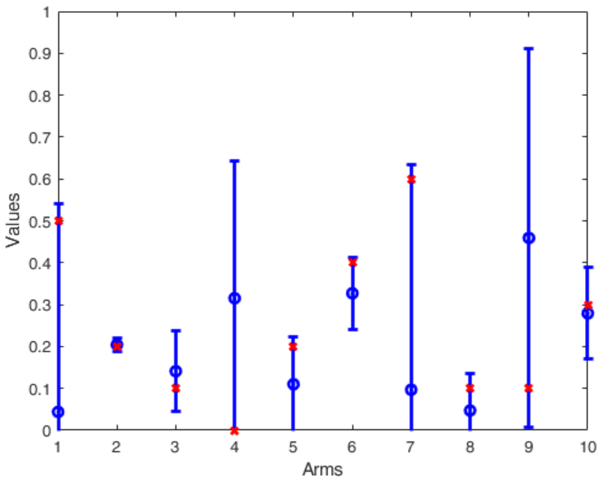

Figure 1: Bounds in a MAB setting.

1. Which arm would a UCB1 algorithm choose for the next round?
2. Do you think that Figure 1 might be the results obtained by running UCB1 for several rounds?
3. Which arm will Thompson Sampling converge to if $T \to \infty$ (starting from a uniform prior)?
4. Assuming that the arms have been pulled $T_1 = 5$, $T_2 = 50$, $T_3 = 10$, $T_4 = 5$, $T_5 = 10$, $T_6 = 10$, $T_7 = 5$, $T_8 = 10$, $T_9 = 2$, and $T_{10} = 10$, compute the pseudo-regret accumulated so far by the used algorithm.

Motivate your answers.

1. The one with the highest upper confidence bound, i.e., arm a_9.
2. No, since UCB1 tends to have all the upper confidence bounds at similar level over its execution.
3. It would converge to the arm with the largest real expected reward, i.e., arm a_7.
4. First, we need to compute the expected reward gaps: $\Delta_1 = 0.6 - 0.5 = 0.1$, $\Delta_2 = 0.4$, $\Delta_3 = 0.5$, $\Delta_4 = 0.6$, $\Delta_5 = 0.4$, $\Delta_6 = 0.2$, $\Delta_8 = 0.5$, $\Delta_9 = 0.5$, $\Delta_{10} = 0.3$.

   As a consequence, the regret is:

$$
\begin{aligned}
R_T &= \mu^* T - \sum_{t=1}^{T} \mu_{i_t} = \sum_{i=1}^{10} \Delta_i T_i \\
&= 0.1 \cdot 5 + 0.4 \cdot 50 + 0.5 \cdot 10 + 0.6 \cdot 5 + 0.4 \cdot 10 + 0.2 \cdot 10 + 0.5 \cdot 10 + 0.5 \cdot 2 + 0.3 \cdot 10 \\
&= 0.5 + 20 + 5 + 3 + 4 + 2 + 5 + 1 + 3 = 43.5,
\end{aligned}
$$

   recalling that $\Delta_7 = 0$ since it is the optimal arm.